

A continuous family of counterexamples to the middle runner conjecture

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Abstract

The middle runner conjecture from the 2015 AIM workshop problem list concerns the time averages of the ordered positions of runners moving at commensurable nonzero speeds on a circular track. As stated, the runners may start at arbitrary positions, and the conjecture asserts that complementary order statistics have averages summing to one. We give a continuous family of counterexamples with three runners. For velocities $(1, 2, 3)$ and initial positions $(0, 0, a)$, let $M(a)$ be the average median position over one common period. We prove the exact formula

$$M(a) = \begin{cases} \frac{1}{2} - \frac{4}{9}a \left(\frac{1}{2} - a \right), & 0 \leq a \leq \frac{1}{2}, \\ \frac{1}{2} + \frac{4}{9} \left(a - \frac{1}{2} \right) (1 - a), & \frac{1}{2} \leq a \leq 1. \end{cases}$$

Consequently, modulo one, the conjectured equality holds in this family only for $a = 0$ and $a = 1/2$; every other offset is a counterexample. In particular, $a = 1/4$ gives ordered-position averages

$$\left(\frac{161}{576}, \frac{17}{36}, \frac{431}{576} \right).$$

The proof is an elementary but complete piecewise integration. Exact rational verification code is provided as supplementary evidence.

1 Introduction

The 2015 AIM workshop problem list on dynamical algebraic combinatorics asks the following question [1, Section 2.1]. Consider n runners moving on a circular track at commensurable nonzero speeds. Positions are normalized to $[0, 1)$, and the runners are explicitly not required to start at the starting line. If

$$p_1(t) \leq p_2(t) \leq \cdots \leq p_n(t)$$

are the ordered positions at time t , let p_i be the average of $p_i(t)$ over one common period. Conjecture 2.1 of [1] states that

$$p_i + p_j = 1 \quad \text{whenever } i + j = n + 1. \quad (1.1)$$

For odd n , the central case says that the average position of the middle runner is $1/2$, which gives the problem its name. The source reports that the equal-speed case had been solved and that the general case remained open in 2015.

This note disproves (1.1) under its published hypotheses. The smallest nontrivial odd case $n = 3$ already contains a one-parameter family for which the average middle position can be evaluated exactly. The parameter space is classified completely: only two offsets modulo one satisfy the proposed symmetry, while all other offsets fail it. Thus the obstruction is not an isolated numerical configuration.

The result is deliberately limited to the fixed velocity triple $(1, 2, 3)$ and the initial positions $(0, 0, a)$. It is not a classification of arbitrary three-runner systems and does not propose a corrected conjecture.

2 Setup and main result

For $x \in \mathbb{R}$, write $\{x\} = x - \lfloor x \rfloor \in [0, 1)$ for its fractional part. Fix $a \in \mathbb{R}/\mathbb{Z}$ and consider the three positions

$$x_1(t) = \{t\}, \quad x_2(t) = \{2t\}, \quad x_3(t) = \{3t + a\}. \quad (2.1)$$

The velocities are positive, nonzero, and commensurable, and one common period is 1. Let

$$m_a(t) = \text{med}(x_1(t), x_2(t), x_3(t))$$

and define

$$M(a) = \int_0^1 m_a(t) dt. \quad (2.2)$$

Changing a by an integer does not change the system, so we use the representative $a \in [0, 1]$.

Theorem 2.1. *For the system in (2.1),*

$$M(a) = \begin{cases} \frac{1}{2} - \frac{4}{9}a \left(\frac{1}{2} - a \right), & 0 \leq a \leq \frac{1}{2}, \\ \frac{1}{2} + \frac{4}{9} \left(a - \frac{1}{2} \right) (1 - a), & \frac{1}{2} \leq a \leq 1. \end{cases} \quad (2.3)$$

Hence $M(a) = 1/2$ exactly when $a \in \{0, 1/2, 1\}$. Equivalently, modulo one, the middle-runner equality holds in this family exactly for $a \in \{0, 1/2\}$.

Corollary 2.2. *For every*

$$a \in (\mathbb{R}/\mathbb{Z}) \setminus \{0, 1/2\},$$

the three-runner system (2.1) is a counterexample to Conjecture 2.1 of [1]. The counterexamples form two open arcs in the parameter circle and therefore an uncountable continuous family.

3 Piecewise evaluation of the median

We first assume

$$0 < a < \frac{1}{2}. \tag{3.1}$$

All changes in the relative order of the three trajectories occur at a wrap point or at an intersection. The internal wrap points of x_3 are

$$A = \frac{1-a}{3}, \quad C = \frac{2-a}{3}, \quad F = 1 - \frac{a}{3},$$

and the internal wrap point of x_2 is $1/2$. The intersections of x_1 with x_3 occur at

$$B = \frac{1-a}{2}, \quad E = 1 - \frac{a}{2},$$

while the only internal intersection of x_2 with x_3 is

$$D = 1 - a.$$

The first two trajectories meet only at the endpoints of the period. Under (3.1), these events have the strict order

$$0 < A < B < \frac{1}{2} < C < D < E < F < 1. \tag{3.2}$$

On each resulting open interval the relative order is fixed. Evaluating the three affine pieces at any interior point gives the following median.

interval	$m_a(t)$
$(0, A)$	$2t$
(A, B)	t
$(B, 1/2)$	$a + 3t - 1$
$(1/2, C)$	t
(C, D)	$2t - 1$
(D, E)	$a + 3t - 2$
(E, F)	t
$(F, 1)$	$2t - 1$

Values at the finitely many intersections, wrap discontinuities, and period endpoints do not affect the integral. Therefore

$$\begin{aligned} M(a) = & A^2 + \frac{B^2 - A^2}{2} + (a-1) \left(\frac{1}{2} - B \right) + \frac{3}{2} \left(\frac{1}{4} - B^2 \right) \\ & + \frac{C^2 - 1/4}{2} + (D^2 - C^2) - (D - C) \\ & + (a-2)(E - D) + \frac{3}{2}(E^2 - D^2) + \frac{F^2 - E^2}{2} + F - F^2. \end{aligned} \quad (3.3)$$

Substituting $A, \dots, F$ and collecting terms yields

$$M(a) = \frac{8a^2 - 4a + 9}{18} = \frac{1}{2} - \frac{4}{9}a \left(\frac{1}{2} - a \right). \quad (3.4)$$

The endpoint values $a = 0$ and $a = 1/2$ follow either by direct subdivision or by dominated convergence, since for all but finitely many t the trajectories and their median converge pointwise as a approaches either endpoint.

For clarity, write $x_i^{(b)}$ for the trajectory in (2.1) with offset parameter b ; the first two trajectories do not depend on b . To obtain the second half of (2.3), compare parameters a and $1 - a$. Away from the finitely many wrap times, time reversal and reflection of the circle give the multiset identity

$$\{x_1^{(a)}(1-t), x_2^{(a)}(1-t), x_3^{(a)}(1-t)\} = \{1 - x_1^{(1-a)}(t), 1 - x_2^{(1-a)}(t), 1 - x_3^{(1-a)}(t)\}.$$

Reflection reverses the order of three numbers but sends their median to one minus the original median. Hence, almost everywhere,

$$m_a(1-t) = 1 - m_{1-a}(t). \quad (3.5)$$

Integration gives

$$M(a) = 1 - M(1-a). \quad (3.6)$$

Using (3.4) for $1 - a \in [0, 1/2]$ proves the second line of (2.3). The two quadratic factors in (2.3) vanish only at $0, 1/2, 1$, completing the proof of Theorem 2.1 and Corollary 2.2.

4 A rational counterexample and the paired equality

The parameter $a = 1/4$ gives a compact exact witness. Integrating all three order statistics over the same subdivision gives

$$(p_1, p_2, p_3) = \left(\frac{161}{576}, \frac{17}{36}, \frac{431}{576} \right). \quad (4.1)$$

In particular,

$$p_2 = \frac{17}{36} \neq \frac{1}{2}, \quad p_1 + p_3 = \frac{37}{36} \neq 1. \quad (4.2)$$

The simultaneous failure in (4.2) is forced throughout the family. Each individual positive-integer-speed trajectory is uniformly distributed over the circle during the common period and therefore has average $1/2$. Pointwise, sorting does not change the sum of the three positions, so

$$p_1 + p_2 + p_3 = \frac{3}{2}. \quad (4.3)$$

Consequently,

$$p_1 + p_3 = \frac{3}{2} - M(a). \quad (4.4)$$

Thus the middle equality $p_2 = 1/2$ and the complementary equality $p_1 + p_3 = 1$ hold or fail together, exactly at the parameters classified in Theorem 2.1.

5 Exact verification

The supplementary Python program `verification/verify_middle_runner_counterexample.py` uses exact rational arithmetic. It independently generates every wrap point and pairwise intersection, orders the affine pieces on each interval, and integrates all three order statistics. It checks:

- the witness $a = 1/4$ in (4.1);
- the common-start positive control $a = 0$;
- invariance under replacing an initial position by an integer translate; and
- formula (2.3) for all 775 distinct rational representatives in $[0, 1]$ with denominator at most 50. Since 0 and 1 represent the same point, these are 774 distinct parameter classes modulo one.

The program uses only the Python standard library and requires Python 3.9 or later. From the directory containing this manuscript and the `verification` subdirectory, run

```
python verification/verify_middle_runner_counterexample.py
```

The finite checks are supplementary evidence and regression tests. They do not prove the statement for all real parameters; the continuous result rests on the piecewise argument in Section 3.

6 Scope and literature boundary

The contribution of this note is the exact evaluation of a complete one-parameter family and the resulting disproof of the middle runner conjecture under the hypotheses printed in [1]. The phrase "as stated" is important only as a matter of scope: arbitrary starting positions are not an inference made here but an explicit part of the published problem statement.

As of 27 August 2026, a targeted search through the exact problem title and wording, combinations of the four problem-list authors with the relevant terminology, and the arXiv, OpenAlex,

and Crossref indexes did not locate a later correction or a directly equivalent counterexample. This is not an exhaustive priority certificate. Unindexed notes, private communications, differently named equivalent results, and later parallel work may exist. Accordingly, this note does not claim absolute global priority, external expert confirmation, or peer review.

The theorem does not classify arbitrary initial phases or arbitrary velocity triples, and it does not settle any strengthened version that imposes a common starting position. Such variants are outside the present claim.

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References

- [1] J. Propp, T. Roby, J. Striker, and N. Williams, *Problems for 2015 AIM workshop on Dynamical Algebraic Combinatorics*, 29 May 2015, Section 2.1, Conjecture 2.1. [AIM problem list](#).